BUSFIN 8200 – Problem Set 2

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Question 1

Question 2

2 (a)

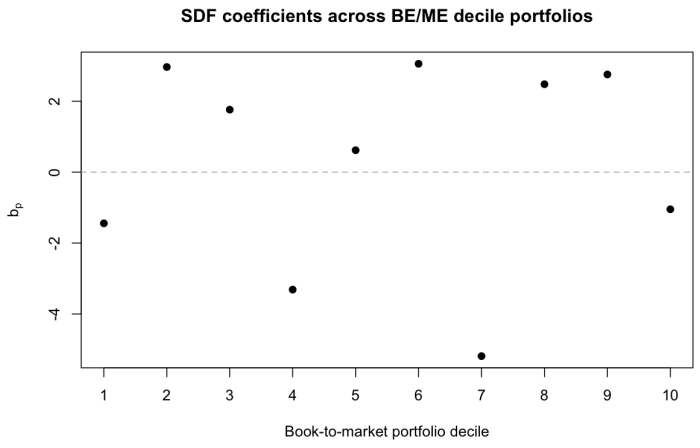
Using the normalization $\mathbb{E}[M^*] = 1$, we find the stochastic discount factor $M_t^* = a - b'xR_t$ that satisfies $\hat{\mathbb{E}}[M^* \cdot xR] = 0$. In particular, we have:

$$\begin{aligned} 0 &= \mathbb{E}[M^* \cdot xR] \\ &= \underbrace{\mathbb{E}[M^*]}_{=1} \cdot \mathbb{E}[xR] + \text{Cov}(M^*, xR) \\ &= \mathbb{E}[xR] - b'\text{Cov}(xR, xR) \\ &\implies \mathbb{E}[xR] = b'\Sigma \\ &\implies b = \Sigma^{-1}\mathbb{E}[xR] \end{aligned}$$

xR is a Tx10 vector of excess returns with T = 1,202 months in the sample from July 1926 to August 2026. Verifying that Σ is invertible, we find that:

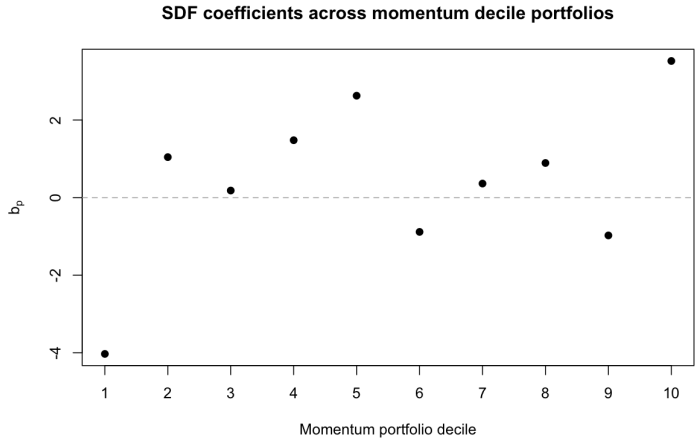
$$\begin{aligned} 1 &= \mathbb{E}[M^*] \\ &= \mathbb{E}[a - b'xR] \\ &= a - b'\mathbb{E}[xR] \\ &\implies a = 1 + b'\mathbb{E}[xR] \end{aligned}$$

Therefore, we are able to compute the stochastic discount factor $M_t^* = a - b'xR_t$ for each month in the sample and obtain the following scatterplot:



2 (b)

Following the same procedure as in part (a), we find the respective b coefficients of the stochastic discount factor that prices the excess returns of the 10 portfolios in the MOM dataset as follows:



Similar to part (a), we do not observe any clear pattern between the b coefficients and the decile portfolios' excess returns. The correlation between the recovered M_t^* is 0.3579, which suggests that the SDF that prices a particular portfolio does not necessarily price the other one as the decile portfolios are sorted using different signals.

2 (c)

2 (d)

2 (e)

Question 3

3 (a)

3 (b)

3 (c)

3 (d)

3 (e)

Question 4

4 (a)

4 (b)

4 (c)

4 (d)

4 (e)